

Introduction to Quantum Technologies and Applications

SITAMS CSE (AI & ML) • R23 • III Year • V Semester • Course Code: 23ESC351T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit 1: INTRODUCTION TO QUANTUM THEORY AND TECHNOLOGIES

Syllabus topics

- The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics – theoretical comparison, Quantum states and measurement: nature of observation, Overview of quantum systems: electrons, photons, atoms, The concept of quantization: discrete energy levels, Why quantum?
- Strategic, scientific, and technological significance, A snapshot of quantum technologies: Computing, Communication, and Sensing, National and global quantum missions: India's Quantum Mission, EU, USA, China

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit 2: THEORETICAL STRUCTURE OF QUANTUM INFORMATION SYSTEMS

Syllabus topics

- What is a qubit?
- Conceptual understanding using spin and polarization, Comparison: classical bits vs quantum bits, Quantum systems: trapped ions, superconducting circuits, photons (nonengineering view), Quantum coherence and decoherence – intuitive explanation, Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract, The role of entanglement and non-locality in systems, Quantum information vs classical information: principles and differences, Philosophical implications: randomness, determinism, and observer role

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Unit 3: BUILDING A QUANTUM COMPUTER – THEORETICAL CHALLENGES AND REQUIREMENTS

Syllabus topics

- What is required to build a quantum computer (conceptual overview)?, Fragility of quantum systems: decoherence, noise, and control, Conditions for a functional quantum system: Isolation, Error management, Scalability, Stability, Theoretical barriers: Why maintaining entanglement is difficult, Error correction as a theoretical necessity, Quantum hardware platforms

(brief conceptual comparison), Superconducting circuits, Trapped ions, Photonics, Vision vs reality: what's working and what remains elusive, The role of quantum software in managing theoretical complexities

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Unit 4: QUANTUM COMMUNICATION AND COMPUTING – THEORETICAL PERSPECTIVE

Syllabus topics

- Quantum vs Classical Information, Basics of Quantum Communication, Quantum Key Distribution (QKD), Role of Entanglement in Communication, The Idea of the Quantum Internet – Secure Global Networking, Introduction to Quantum Computing, Quantum Parallelism (Many States at Once), Classical vs Quantum Gates, Challenges: Decoherence and Error Correction, Real-World Importance and Future Potential

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Unit 5: APPLICATIONS, USE CASES, AND THE QUANTUM FUTURE

Syllabus topics

- Real-world application domains: Healthcare (drug discovery), Material science, Logistics and optimization, Quantum sensing and precision timing, Industrial case studies: IBM, Google, Microsoft, PsiQuantum, Ethical, societal, and policy considerations, Challenges to adoption: cost, skills, standardization, Emerging careers in quantum: roles, skillsets, and preparation pathways, Educational and research landscape – India's opportunity in the global quantum race

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